

MATH I – Week 2

Fall 2026

Functions

OpenStax Calculus 1

1

FUNCTIONS AND GRAPHS

Figure 1.1 A portion of the San Andreas Fault in California. Major faults like this are the sites of most of the strongest earthquakes ever recorded. (credit: modification of work by Robb Hannawacker, NPS)

Chapter Outline

- 1.1 Review of Functions
- 1.2 Basic Classes of Functions
- 1.3 Trigonometric Functions
- 1.4 Inverse Functions
- 1.5 Exponential and Logarithmic Functions



Introduction

In the past few years, major earthquakes have occurred in several countries around the world. In January 2010, an earthquake of magnitude 7.3 hit Haiti. A magnitude 9 earthquake shook northeastern Japan in March 2011. In April 2014, an 8.2-magnitude earthquake struck off the coast of northern Chile. What do these numbers mean? In particular, how does a magnitude 9 earthquake compare with an earthquake of magnitude 8.2? Or 7.3? Later in this chapter, we show how logarithmic functions are used to compare the relative intensity of two earthquakes based on the magnitude of each earthquake (see [Example 1.39](#)).

Calculus is the mathematics that describes changes in functions. In this chapter, we review all the functions necessary to study calculus. We define polynomial, rational, trigonometric, exponential, and logarithmic functions. We review how to evaluate these functions, and we show the properties of their graphs. We provide examples of equations with terms involving these functions and illustrate the algebraic techniques necessary to solve them. In short, this chapter provides the foundation for the material to come. It is essential to be familiar and comfortable with these ideas before proceeding to the formal introduction of calculus in the next chapter.

Functions

Definition

A **function** f consists of a set of inputs, a set of outputs, and a rule for assigning each input to exactly one output. The set of inputs is called the **domain** of the function. The set of outputs is called the **range** of the function.

For a general function f with domain D , we often use x to denote the input and y to denote the output associated with x . When doing so, we refer to x as the **independent variable** and y as the **dependent variable**, because it depends on x . Using function notation, we write $y = f(x)$, and we read this equation as “ y equals f of x .” For the squaring function described earlier, we write $f(x) = x^2$.

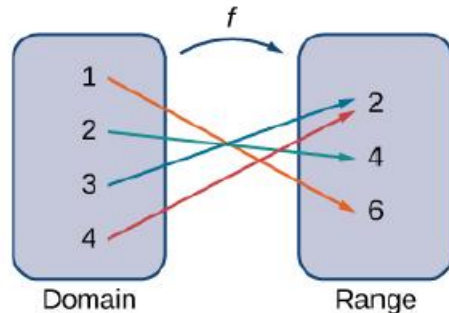


Figure 1.3 A function maps every element in the domain to exactly one element in the range. Although only one output is sent to, two different inputs can be sent to the same output.

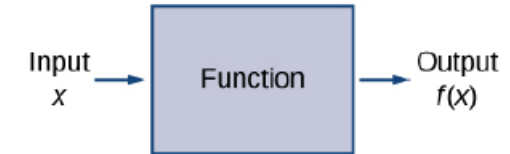


Figure 1.2 A function can be visualized as an input/output device.

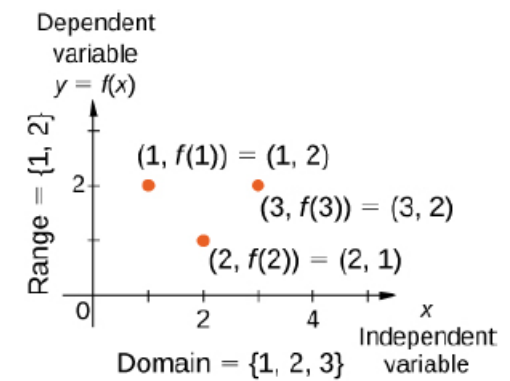


Figure 1.4 In this case, a graph of a function f has a domain of $\{1, 2, 3\}$ and a range of $\{1, 2\}$.

In-Class activity

Evaluating Functions

For the function $f(x) = 3x^2 + 2x - 1$, evaluate

- a. $f(-2)$
- b. $f(\sqrt{2})$
- c. $f(a + h)$

In-Class activity *solution*

Evaluating Functions

For the function $f(x) = 3x^2 + 2x - 1$, evaluate

- a. $f(-2)$
- b. $f(\sqrt{2})$
- c. $f(a+h)$

✓ Solution

Substitute the given value for x in the formula for $f(x)$.

a. $f(-2) = 3(-2)^2 + 2(-2) - 1 = 12 - 4 - 1 = 7$

b. $f(\sqrt{2}) = 3(\sqrt{2})^2 + 2\sqrt{2} - 1 = 6 + 2\sqrt{2} - 1 = 5 + 2\sqrt{2}$

c.
$$\begin{aligned} f(a+h) &= 3(a+h)^2 + 2(a+h) - 1 = 3(a^2 + 2ah + h^2) + 2a + 2h - 1 \\ &= 3a^2 + 6ah + 3h^2 + 2a + 2h - 1 \end{aligned}$$

Notation in textbook: intervals

$$(1, 5) = \{x | 1 < x < 5\}.$$

The numbers 1 and 5 are called the *endpoints* of this set. If we want to consider the set that includes the endpoints, we would denote this set by writing

$$[1, 5] = \{x | 1 \leq x \leq 5\}.$$

We can use similar notation if we want to include one of the endpoints, but not the other. To denote the set of nonnegative real numbers, we would use the set-builder notation

$$\{x | 0 \leq x\}.$$

The smallest number in this set is zero, but this set does not have a largest number. Using interval notation, we would use the symbol ∞ , which refers to positive infinity, and we would write the set as

$$[0, \infty) = \left\{x \mid 0 \leq x\right\}.$$

It is important to note that ∞ is not a real number. It is used symbolically here to indicate that this set includes all real numbers greater than or equal to zero. Similarly, if we wanted to describe the set of all nonpositive numbers, we could write

$$(-\infty, 0] = \left\{x \mid x \leq 0\right\}.$$

Here, the notation $-\infty$ refers to negative infinity, and it indicates that we are including all numbers less than or equal to zero, no matter how small. The set

$$(-\infty, \infty) = \left\{x \mid x \text{ is any real number}\right\}$$

refers to the set of all real numbers.

In-Class activity

solve c.

Finding Domain and Range

For each of the following functions, determine the i. domain and ii. range.

a. $f(x) = (x - 4)^2 + 5$

b. $f(x) = \sqrt{3x + 2} - 1$

c. $f(x) = \frac{3}{x-2}$

In-Class activity *solution (c.)*

Finding Domain and Range

For each of the following functions, determine the i. domain and ii. range.

a. $f(x) = (x - 4)^2 + 5$

b. $f(x) = \sqrt{3x + 2} - 1$

c. $f(x) = \frac{3}{x-2}$

c. Consider $f(x) = 3/(x - 2)$.

i. Since $3/(x - 2)$ is defined when the denominator is nonzero, the domain is $\{x|x \neq 2\}$.

ii. To find the range of f , we need to find the values of y such that there exists a real number x in the domain with the property that

$$\frac{3}{x-2} = y.$$

Solving this equation for x , we find that

$$x = \frac{3}{y} + 2.$$

Therefore, as long as $y \neq 0$, there exists a real number x in the domain such that $f(x) = y$. Thus, the range is $\{y|y \neq 0\}$.

Representing Functions

Typically, a function is represented using one or more of the following tools:

- A table
- A graph
- A formula

Hours after Midnight	Temperature (°F)	Hours after Midnight	Temperature (°F)
0	58	12	84
1	54	13	85
2	53	14	85
3	52	15	83
4	52	16	82
5	55	17	80
6	60	18	77
7	64	19	74
8	72	20	69
9	75	21	65
10	78	22	60
11	80	23	58

Table 1.1 Temperature as a Function of Time of Day

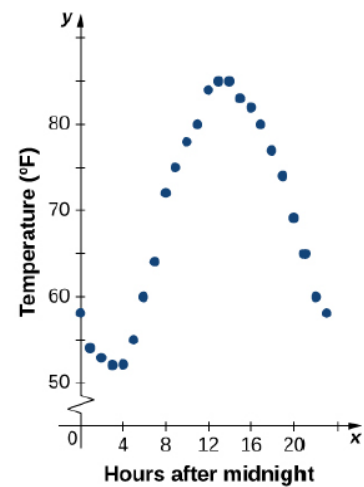


Figure 1.6 The graph of the data from Table 1.1 shows temperature as a function of time.

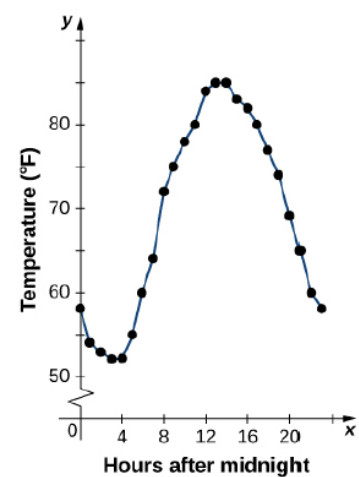
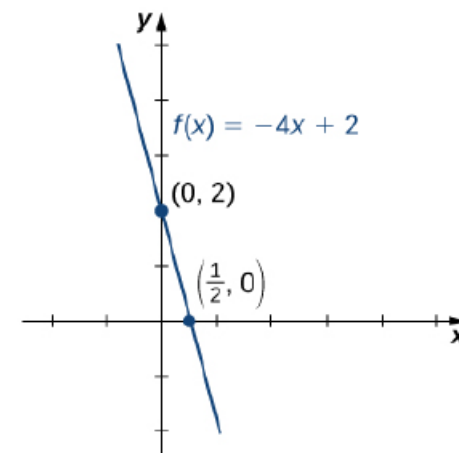


Figure 1.7 Connecting the dots in Figure 1.6 shows the general pattern of the data.

When creating a table of inputs and outputs, we typically check to determine whether zero is an output. Those values of x where $f(x) = 0$ are called the **zeros of a function**. For example, the zeros of $f(x) = x^2 - 4$ are $x = \pm 2$. The zeros determine where the graph of f intersects the x -axis, which gives us more information about the shape of the graph of the function. The graph of a function may never intersect the x -axis, or it may intersect multiple (or even infinitely many) times.

Another point of interest is the y -intercept, if it exists. The y -intercept is given by $(0, f(0))$.

Since a function has exactly one output for each input, the graph of a function can have, at most, one y -intercept. If $x = 0$ is in the domain of a function f , then f has exactly one y -intercept. If $x = 0$ is not in the domain of f , then f has no y -intercept. Similarly, for any real number c , if c is in the domain of f , there is exactly one output $f(c)$, and the line $x = c$ intersects the graph of f exactly once. On the other hand, if c is not in the domain of f , $f(c)$ is not defined and the line $x = c$ does not intersect the graph of f . This property is summarized in the **vertical line test**.



Rule: Vertical Line Test

Given a function f , every vertical line that may be drawn intersects the graph of f no more than once. If any vertical line intersects a set of points more than once, the set of points does not represent a function.

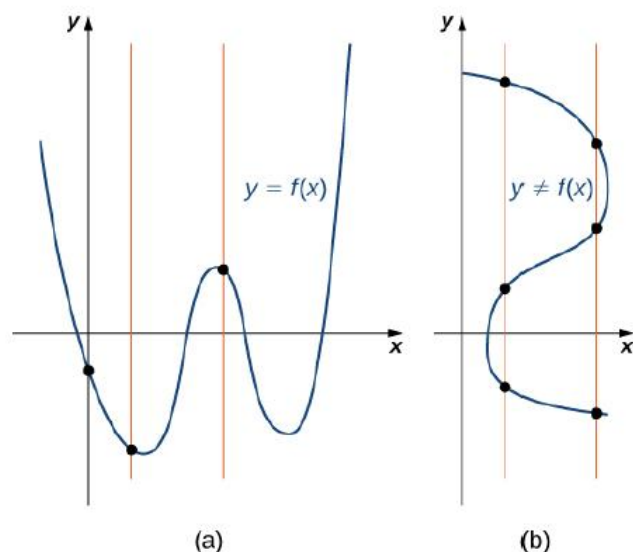


Figure 1.8 (a) The set of plotted points represents the graph of a function because every vertical line intersects the set of points, at most, once. (b) The set of plotted points does not represent the graph of a function because some vertical lines intersect the set of points more than once.

Zeros, intercept,
vertical line test

Finding Zeros and y -Intercepts of a Function

Consider the function $f(x) = -4x + 2$.

- Find all zeros of f .
- Find the y -intercept (if any).
- Sketch a graph of f .

EXAMPLE 1.3

Finding Zeros and y -Intercepts of a Function

Consider the function $f(x) = -4x + 2$.

- Find all zeros of f .
- Find the y -intercept (if any).
- Sketch a graph of f .

✓ Solution

- To find the zeros, solve $f(x) = -4x + 2 = 0$. We discover that f has one zero at $x = 1/2$.
- The y -intercept is given by $(0, f(0)) = (0, 2)$.
- Given that f is a linear function of the form $f(x) = mx + b$ that passes through the points $(1/2, 0)$ and $(0, 2)$, we can sketch the graph of f (Figure 1.9).

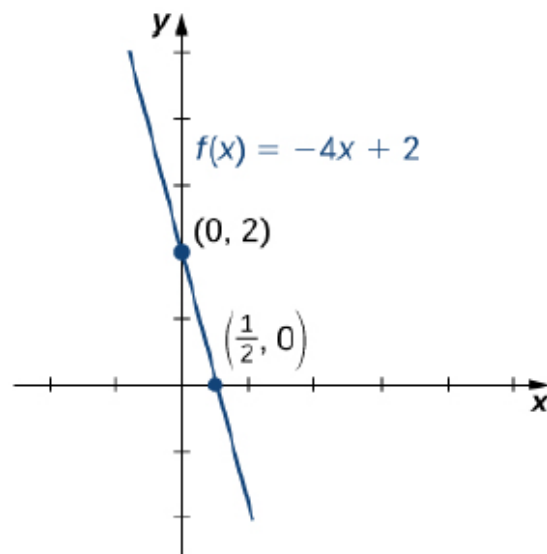


Figure 1.9 The function $f(x) = -4x + 2$ is a line with x -intercept $(1/2, 0)$ and y -intercept $(0, 2)$.

Solution

Increasing or decreasing on an Interval

Definition

We say that a function f is **increasing on the interval** I if for all $x_1, x_2 \in I$,

$$f(x_1) \leq f(x_2) \text{ when } x_1 < x_2.$$

We say f is strictly increasing on the interval I if for all $x_1, x_2 \in I$,

$$f(x_1) < f(x_2) \text{ when } x_1 < x_2.$$

We say that a function f is **decreasing on the interval** I if for all $x_1, x_2 \in I$,

$$f(x_1) \geq f(x_2) \text{ if } x_1 < x_2.$$

We say that a function f is strictly decreasing on the interval I if for all $x_1, x_2 \in I$,

$$f(x_1) > f(x_2) \text{ if } x_1 < x_2.$$

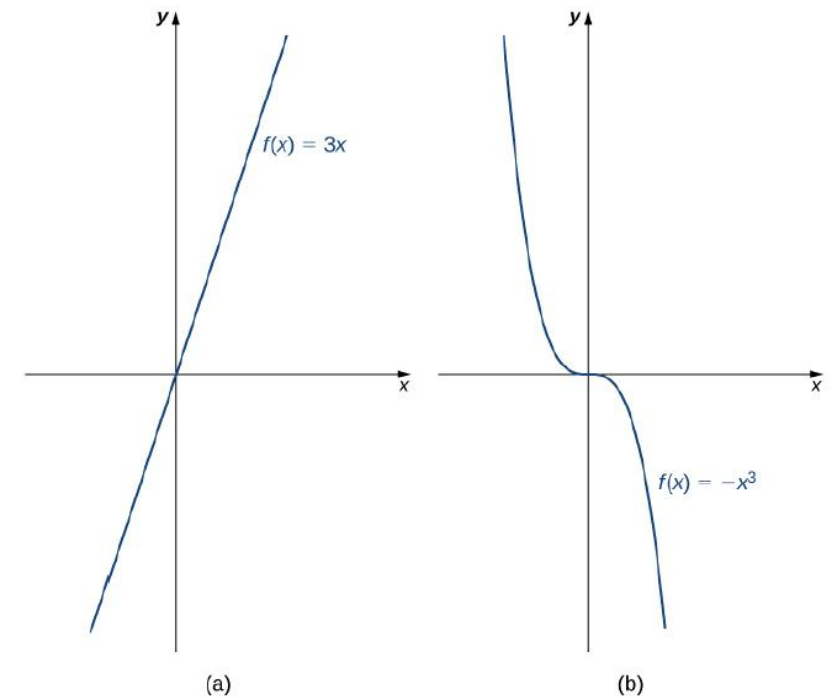


Figure 1.11 (a) The function $f(x) = 3x$ is increasing on the interval $(-\infty, \infty)$. (b) The function $f(x) = -x^3$ is decreasing on the interval $(-\infty, \infty)$.

Combining functions

Combining Functions with Mathematical Operators

To combine functions using mathematical operators, we simply write the functions with the operator and simplify. Given two functions f and g , we can define four new functions:

$(f + g)(x) = f(x) + g(x)$	<i>Sum</i>
$(f - g)(x) = f(x) - g(x)$	<i>Difference</i>
$(f \cdot g)(x) = f(x)g(x)$	<i>Product</i>
$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ for } g(x) \neq 0$	<i>Quotient</i>

Combining Functions Using Mathematical Operations

Given the functions $f(x) = 2x - 3$ and $g(x) = x^2 - 1$, find each of the following functions and state its domain.

- a. $(f + g)(x)$
- b. $(f - g)(x)$
- c. $(f \cdot g)(x)$
- d. $\left(\frac{f}{g}\right)(x)$

✓ Solution

- a. $(f + g)(x) = (2x - 3) + (x^2 - 1) = x^2 + 2x - 4$. The domain of this function is the interval $(-\infty, \infty)$.
- b. $(f - g)(x) = (2x - 3) - (x^2 - 1) = -x^2 + 2x - 2$. The domain of this function is the interval $(-\infty, \infty)$.
- c. $(f \cdot g)(x) = (2x - 3)(x^2 - 1) = 2x^3 - 3x^2 - 2x + 3$. The domain of this function is the interval $(-\infty, \infty)$.
- d. $\left(\frac{f}{g}\right)(x) = \frac{2x-3}{x^2-1}$. The domain of this function is $\{x \mid x \neq \pm 1\}$.

Function composition

Definition

Consider the function f with domain A and range B , and the function g with domain D and range E . If B is a subset of D , then the **composite function** $(g \circ f)(x)$ is the function with domain A such that

$$(g \circ f)(x) = g(f(x)). \quad (1.1)$$

Compositions of Functions Defined by Formulas

Consider the functions $f(x) = x^2 + 1$ and $g(x) = 1/x$.

- a. Find $(g \circ f)(x)$ and state its domain and range.

✓ Solution

- a. We can find the formula for $(g \circ f)(x)$ in two different ways. We could write

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = \frac{1}{x^2 + 1}.$$

Alternatively, we could write

$$(g \circ f)(x) = g(f(x)) = \frac{1}{f(x)} = \frac{1}{x^2 + 1}.$$

EXAMPLE 1.7**Compositions of Functions Defined by Formulas**

Consider the functions $f(x) = x^2 + 1$ and $g(x) = 1/x$.

- Find $(g \circ f)(x)$ and state its domain and range.
- Evaluate $(g \circ f)(4)$, $(g \circ f)(-1/2)$.
- Find $(f \circ g)(x)$ and state its domain and range.
- Evaluate $(f \circ g)(4)$, $(f \circ g)(-1/2)$.

✓ Solution

- We can find the formula for $(g \circ f)(x)$ in two different ways. We could write

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = \frac{1}{x^2 + 1}.$$

Alternatively, we could write

$$(g \circ f)(x) = g(f(x)) = \frac{1}{f(x)} = \frac{1}{x^2 + 1}.$$

Since $x^2 + 1 \neq 0$ for all real numbers x , the domain of $(g \circ f)(x)$ is the set of all real numbers. Since $0 < 1/(x^2 + 1) \leq 1$, the range is, at most, the interval $(0, 1]$. To show that the range is this entire interval, we let $y = 1/(x^2 + 1)$ and solve this equation for x to show that for all y in the interval $(0, 1]$, there exists a real number x such that $y = 1/(x^2 + 1)$. Solving this equation for x , we see that $x^2 + 1 = 1/y$, which implies that

$$x = \pm \sqrt{\frac{1}{y} - 1}.$$

If y is in the interval $(0, 1]$, the expression under the radical is nonnegative, and therefore there exists a real number x such that $1/(x^2 + 1) = y$. We conclude that the range of $g \circ f$ is the interval $(0, 1]$.

- $(g \circ f)(4) = g(f(4)) = g(4^2 + 1) = g(17) = \frac{1}{17}$
 $(g \circ f)(-1/2) = g(f(-1/2)) = g\left(\left(-\frac{1}{2}\right)^2 + 1\right) = g\left(\frac{5}{4}\right) = \frac{4}{5}$
- We can find a formula for $(f \circ g)(x)$ in two ways. First, we could write

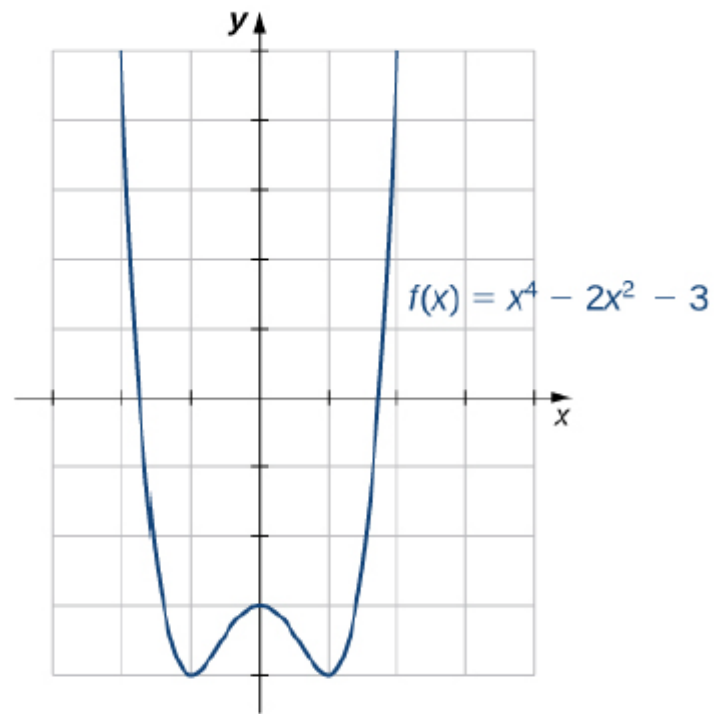
$$(f \circ g)(x) = f(g(x)) = f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^2 + 1.$$

Alternatively, we could write

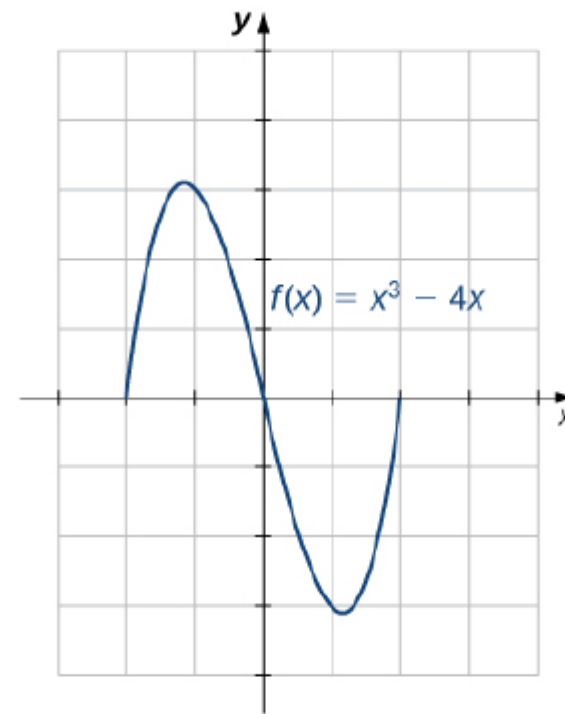
$$(f \circ g)(x) = f(g(x)) = (g(x))^2 + 1 = \left(\frac{1}{x}\right)^2 + 1.$$

$$\begin{aligned} \text{d. } (f \circ g)(4) &= f(g(4)) = f\left(\frac{1}{4}\right) = \left(\frac{1}{4}\right)^2 + 1 = \frac{17}{16} \\ (f \circ g)\left(-\frac{1}{2}\right) &= f\left(g\left(-\frac{1}{2}\right)\right) = f(-2) = (-2)^2 + 1 = 5 \end{aligned}$$

Symmetry of functions



(a) Symmetry about the y -axis



(b) Symmetry about the origin

Figure 1.13 (a) A graph that is symmetric about the y -axis. (b) A graph that is symmetric about the origin.

What about function $|x|$?

One symmetric function that arises frequently is the **absolute value function**, written as $|x|$. The absolute value function is defined as

$$f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases} \quad (1.2)$$

Some students describe this function by stating that it “makes everything positive.” By the definition of the absolute value function, we see that if $x < 0$, then $|x| = -x > 0$, and if $x > 0$, then $|x| = x > 0$. However, for $x = 0$, $|x| = 0$. Therefore, it is more accurate to say that for all nonzero inputs, the output is positive, but if $x = 0$, the output $|x| = 0$. We conclude that the range of the absolute value function is $\{y | y \geq 0\}$. In [Figure 1.14](#), we see that the absolute value function is symmetric about the y -axis and is therefore an even function.

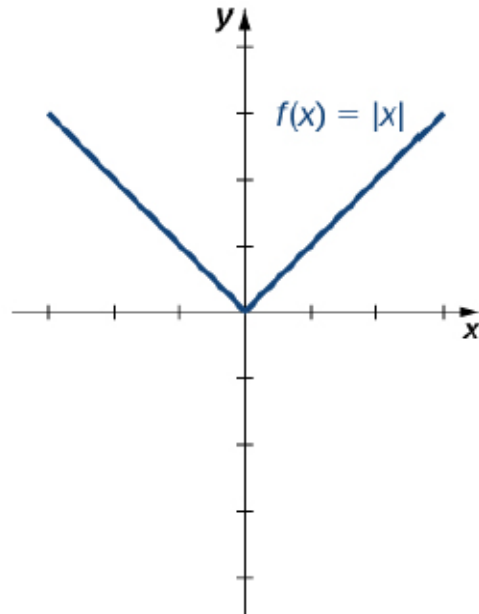


Figure 1.14 The graph of $f(x) = |x|$ is symmetric about the y -axis.

Even and Odd functions & example

Definition

If $f(x) = f(-x)$ for all x in the domain of f , then f is an **even function**. An even function is symmetric about the y -axis.

If $f(-x) = -f(x)$ for all x in the domain of f , then f is an **odd function**. An odd function is symmetric about the origin.

Even and Odd Functions

Determine whether each of the following functions is even, odd, or neither.

- a. $f(x) = -5x^4 + 7x^2 - 2$
- b. $f(x) = 2x^5 - 4x + 5$
- c. $f(x) = \frac{3x}{x^2+1}$

✓ Solution

To determine whether a function is even or odd, we evaluate $f(-x)$ and compare it to $f(x)$ and $-f(x)$.

- a. $f(-x) = -5(-x)^4 + 7(-x)^2 - 2 = -5x^4 + 7x^2 - 2 = f(x)$. Therefore, f is even.
- b. $f(-x) = 2(-x)^5 - 4(-x) + 5 = -2x^5 + 4x + 5$. Now, $f(-x) \neq f(x)$. Furthermore, noting that $-f(x) = -2x^5 + 4x - 5$, we see that $f(-x) \neq -f(x)$. Therefore, f is neither even nor odd.
- c. $f(-x) = 3(-x)/((-x)^2 + 1) = -3x/(x^2 + 1) = -[3x/(x^2 + 1)] = -f(x)$. Therefore, f is odd.

Basic classes of functions

a. Linear functions and slope

Linear Functions and Slope

The easiest type of function to consider is a **linear function**. Linear functions have the form $f(x) = ax + b$, where a and b are constants. In [Figure 1.15](#), we see examples of linear functions when a is positive, negative, and zero. Note that if $a > 0$, the graph of the line rises as x increases. In other words, $f(x) = ax + b$ is increasing on $(-\infty, \infty)$. If $a < 0$, the graph of the line falls as x increases. In this case, $f(x) = ax + b$ is decreasing on $(-\infty, \infty)$. If $a = 0$, the line is horizontal.

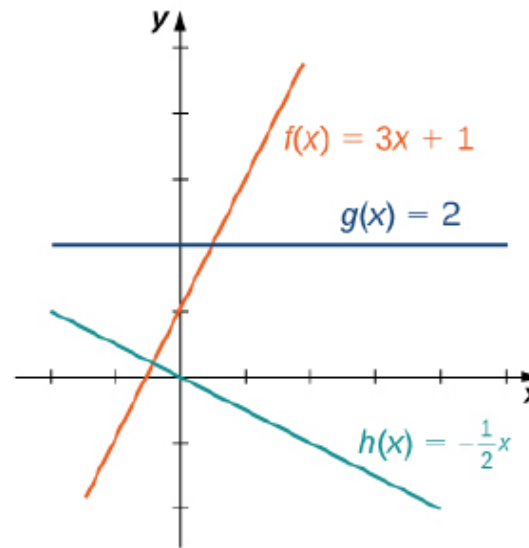
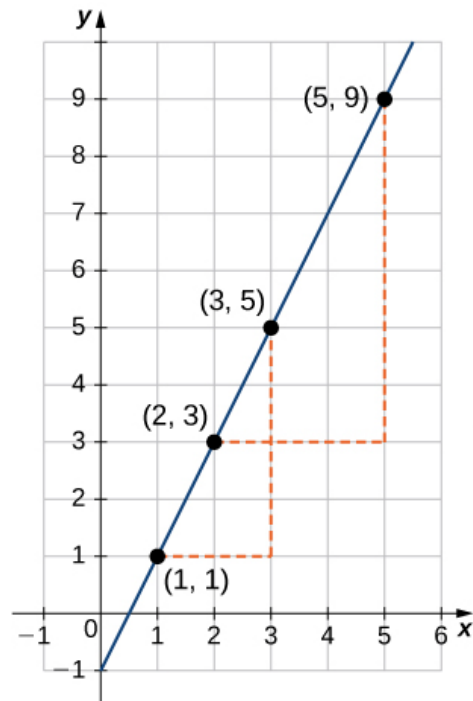


Figure 1.15 These linear functions are increasing or decreasing on $(-\infty, \infty)$ and one function is a horizontal line.

Linear functions and slope



$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{9 - 3}{5 - 2} = 2$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 3}{3 - 2} = 2$$

Definition

Consider a line passing through the point (x_1, y_1) with slope m . The equation

$$y - y_1 = m(x - x_1) \quad (1.4)$$

is the **point-slope equation** for that line.

Consider a line with slope m and y -intercept $(0, b)$. The equation

$$y = mx + b \quad (1.5)$$

is an equation for that line in **slope-intercept form**.

The **standard form of a line** is given by the equation

$$ax + by = c, \quad (1.6)$$

where a and b are not both zero. This form is more general because it allows for a vertical line, $x = k$.

Figure 1.16 For any linear function, the slope $(y_2 - y_1)/(x_2 - x_1)$ is independent of the choice of points (x_1, y_1) and (x_2, y_2) on the line.

Definition

Consider line L passing through points (x_1, y_1) and (x_2, y_2) . Let $\Delta y = y_2 - y_1$ and $\Delta x = x_2 - x_1$ denote the changes in y and x , respectively. The **slope** of the line is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}. \quad (1.3)$$

Polynomials

A linear function is a special type of a more general class of functions: polynomials. A **polynomial function** is any function that can be written in the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \quad (1.7)$$

for some integer $n \geq 0$ and constants $a_n, a_{n-1}, \dots, a_0$, where $a_n \neq 0$. In the case when $n = 0$, we allow for $a_0 = 0$; if $a_0 = 0$, the function $f(x) = 0$ is called the *zero function*. The value n is called the **degree** of the polynomial; the constant a_n is called the *leading coefficient*. A linear function of the form $f(x) = mx + b$ is a polynomial of degree 1 if $m \neq 0$ and degree 0 if $m = 0$. A polynomial of degree 0 is also called a *constant function*. A polynomial function of degree 2 is called a **quadratic function**. In particular, a quadratic function has the form $f(x) = ax^2 + bx + c$, where $a \neq 0$. A polynomial function of degree 3 is called a **cubic function**.

Power Functions

Some polynomial functions are power functions. A **power function** is any function of the form $f(x) = ax^b$, where a and b are any real numbers. The exponent in a power function can be any real number, but here we consider the case when the exponent is a positive integer. (We consider other cases later.) If the exponent is a positive integer, then $f(x) = ax^n$ is a polynomial. If n is even, then $f(x) = ax^n$ is an even function because $f(-x) = a(-x)^n = ax^n$ if n is even. If n is odd, then $f(x) = ax^n$ is an odd function because $f(-x) = a(-x)^n = -ax^n$ if n is odd ([Figure 1.18](#)).

b. Polynomials
c. Power functions

Zeros of Polynomial Functions

Another characteristic of the graph of a polynomial function is where it intersects the x -axis. To determine where a function f intersects the x -axis, we need to solve the equation $f(x) = 0$ for x . In the case of the linear function $f(x) = mx + b$, the x -intercept is given by solving the equation $mx + b = 0$. In this case, we see that the x -intercept is given by $(-b/m, 0)$. In the case of a quadratic function, finding the x -intercept(s) requires finding the zeros of a quadratic equation: $ax^2 + bx + c = 0$. In some cases, it is easy to factor the polynomial $ax^2 + bx + c$ to find the zeros. If not, we make use of the quadratic formula.

Rule: The Quadratic Formula

Consider the quadratic equation

$$ax^2 + bx + c = 0,$$

where $a \neq 0$. The solutions of this equation are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (1.8)$$

If the discriminant $b^2 - 4ac > 0$, this formula tells us there are two real numbers that satisfy the quadratic equation. If $b^2 - 4ac = 0$, this formula tells us there is only one solution, and it is a real number. If $b^2 - 4ac < 0$, no real numbers satisfy the quadratic equation.

Mathematical Models

A large variety of real-world situations can be described using **mathematical models**. A mathematical model is a method of simulating real-life situations with mathematical equations. Physicists, engineers, economists, and other researchers develop models by combining observation with quantitative data to develop equations, functions, graphs, and other mathematical tools to describe the behavior of various systems accurately. Models are useful because they help predict future outcomes. Examples of mathematical models include the study of population dynamics, investigations of weather patterns, and predictions of product sales.

Algebraic VS Transcendental functions

Algebraic Functions

By allowing for quotients and fractional powers in polynomial functions, we create a larger class of functions. An **algebraic function** is one that involves addition, subtraction, multiplication, division, rational powers, and roots. Two types of algebraic functions are rational functions and root functions.

Just as rational numbers are quotients of integers, rational functions are quotients of polynomials. In particular, a **rational function** is any function of the form $f(x) = p(x)/q(x)$, where $p(x)$ and $q(x)$ are polynomials. For example,

$$f(x) = \frac{3x - 1}{5x + 2} \quad \text{and} \quad g(x) = \frac{4}{x^2 + 1}$$

are rational functions. A **root function** is a power function of the form $f(x) = x^{1/n}$, where n is a positive integer greater than one. For example, $f(x) = x^{1/2} = \sqrt{x}$ is the square-root function and $g(x) = x^{1/3} = \sqrt[3]{x}$ is the cube-root function.

By allowing for compositions of root functions and rational functions, we can create other algebraic functions. For example, $f(x) = \sqrt{4 - x^2}$ is an algebraic function.

Transcendental Functions

Thus far, we have discussed algebraic functions. Some functions, however, cannot be described by basic algebraic operations. These functions are known as **transcendental functions** because they are said to “transcend,” or go beyond, algebra. The most common transcendental functions are trigonometric, exponential, and logarithmic functions. A *trigonometric function* relates the ratios of two sides of a right triangle. They are $\sin x$, $\cos x$, $\tan x$, $\cot x$, $\sec x$, and $\csc x$. (We discuss trigonometric functions later in the chapter.) An exponential function is a function of the form $f(x) = b^x$, where the base $b > 0$, $b \neq 1$. A **logarithmic function** is a function of the form $f(x) = \log_b(x)$ for some constant $b > 0$, $b \neq 1$, where $\log_b(x) = y$ if and only if $b^y = x$. (We also discuss exponential and logarithmic functions later in the chapter.)

Independent Study Assignment

Study examples 1.16 and 1.17, which are about domain and range of algebraic functions.

Study example 1.18 on classifying algebraic and transcendental functions.

Piecewise defined Functions

& In-Class activity

Piecewise-Defined Functions

Sometimes a function is defined by different formulas on different parts of its domain. A function with this property is known as a **piecewise-defined function**. The absolute value function is an example of a piecewise-defined function because the formula changes with the sign of x :

$$f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}.$$

Other piecewise-defined functions may be represented by completely different formulas, depending on the part of the domain in which a point falls. To graph a piecewise-defined function, we graph each part of the function in its respective domain, on the same coordinate system. If the formula for a function is different for $x < a$ and $x > a$, we need to pay special attention to what happens at $x = a$ when we graph the function. Sometimes the graph needs to include an open or closed circle to indicate the value of the function at $x = a$. We examine this in the next example.

EXAMPLE 1.19

Graphing a Piecewise-Defined Function

Sketch a graph of the following piecewise-defined function:

$$f(x) = \begin{cases} x + 3, & x < 1 \\ (x - 2)^2, & x \geq 1 \end{cases}.$$

Solution

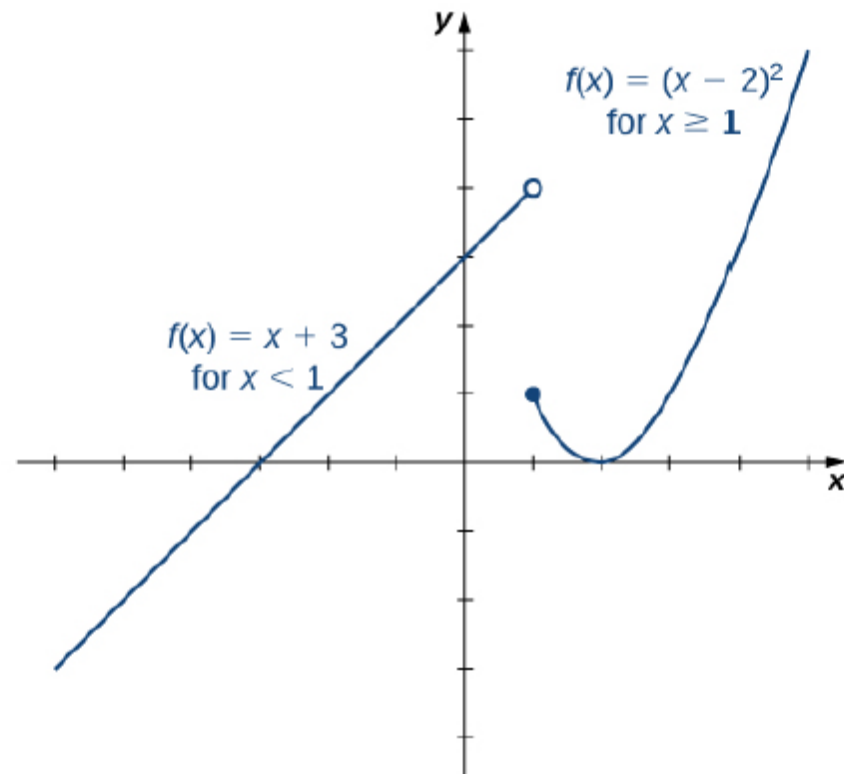


Figure 1.22 This piecewise-defined function is linear for $x < 1$ and quadratic for $x \geq 1$.

EXAMPLE 1.20

Parking Fees Described by a Piecewise-Defined Function

In a big city, drivers are charged variable rates for parking in a parking garage. They are charged \$10 for the first hour or any part of the first hour and an additional \$2 for each hour or part thereof up to a maximum of \$30 for the day. The parking garage is open from 6 a.m. to 12 midnight.

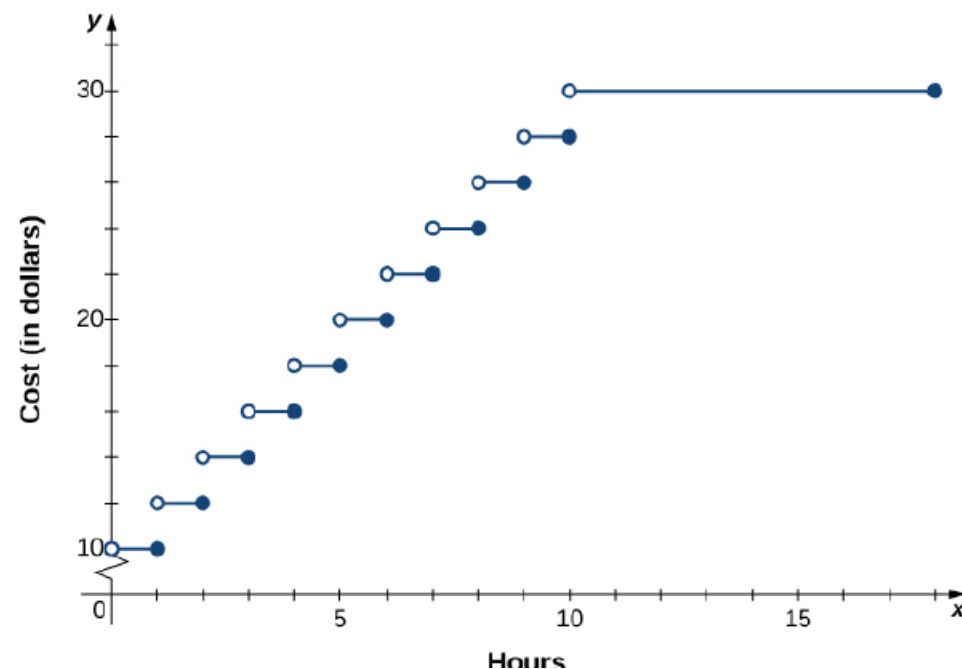
- Write a piecewise-defined function that describes the cost C to park in the parking garage as a function of hours parked x .
- Sketch a graph of this function $C(x)$.

✓ Solution

- Since the parking garage is open 18 hours each day, the domain for this function is $\{x | 0 < x \leq 18\}$. The cost to park a car at this parking garage can be described piecewise by the function

$$C(x) = \begin{cases} 10, & 0 < x \leq 1 \\ 12, & 1 < x \leq 2 \\ 14, & 2 < x \leq 3 \\ 16, & 3 < x \leq 4 \\ \vdots & \\ 30, & 10 < x \leq 18 \end{cases}$$

- The graph of the function consists of several horizontal line segments.



Example

Transformations of Functions

We have seen several cases in which we have added, subtracted, or multiplied constants to form variations of simple functions. In the previous example, for instance, we subtracted 2 from the argument of the function $y = x^2$ to get the function $f(x) = (x - 2)^2$. This subtraction represents a shift of the function $y = x^2$ two units to the right. A shift, horizontally or vertically, is a type of **transformation of a function**. Other transformations include horizontal and vertical scalings, and reflections about the axes.

Transformation of f ($c > 0$)	Effect on the graph of f
$f(x) + c$	Vertical shift up c units
$f(x) - c$	Vertical shift down c units
$f(x + c)$	Shift left by c units
$f(x - c)$	Shift right by c units
$cf(x)$	Vertical stretch if $c > 1$; vertical compression if $0 < c < 1$
$f(cx)$	Horizontal stretch if $0 < c < 1$; horizontal compression if $c > 1$
$-f(x)$	Reflection about the x -axis
$f(-x)$	Reflection about the y -axis

Table 1.7 Transformations of Functions

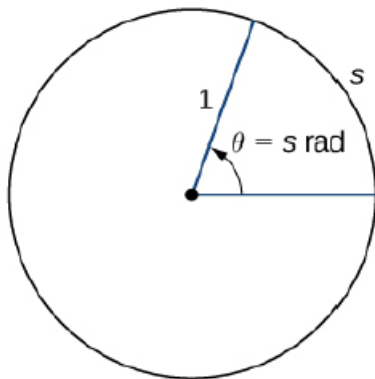


Figure 1.30 The radian measure of an angle θ is the arc length s of the associated arc on the unit circle.

Degrees	Radians	Degrees	Radians
0	0	120	$2\pi/3$
30	$\pi/6$	135	$3\pi/4$
45	$\pi/4$	150	$5\pi/6$
60	$\pi/3$	180	π
90	$\pi/2$		

Table 1.8 Common Angles Expressed in Degrees and Radians

EXAMPLE 1.22

Converting between Radians and Degrees

- Express 225° using radians.
- Express $5\pi/3$ rad using degrees.

Solution

Use the fact that 180° is equivalent to π radians as a conversion factor: $1 = \frac{\pi \text{ rad}}{180^\circ} = \frac{180^\circ}{\pi \text{ rad}}$.

- $225^\circ = 225^\circ \cdot \frac{\pi}{180^\circ} = \frac{5\pi}{4} \text{ rad}$
- $\frac{5\pi}{3} \text{ rad} = \frac{5\pi}{3} \cdot \frac{180^\circ}{\pi} = 300^\circ$

Trigonometric functions

Definition

Let $P = (x, y)$ be a point on the unit circle centered at the origin O . Let θ be an angle with an initial side along the positive x -axis and a terminal side given by the line segment OP . The **trigonometric functions** are then defined as

$$\begin{aligned} \sin \theta &= y & \csc \theta &= \frac{1}{y} \\ \cos \theta &= x & \sec \theta &= \frac{1}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y} \end{aligned} \quad (1.9)$$

If $x = 0$, $\sec \theta$ and $\tan \theta$ are undefined. If $y = 0$, then $\cot \theta$ and $\csc \theta$ are undefined.

We can see that for a point $P = (x, y)$ on a circle of radius r with a corresponding angle θ , the coordinates x and y satisfy

$$\begin{aligned} \cos \theta &= \frac{x}{r} \\ x &= r \cos \theta \\ \sin \theta &= \frac{y}{r} \\ y &= r \sin \theta. \end{aligned}$$

The values of the other trigonometric functions can be expressed in terms of x , y , and r (Figure 1.32).

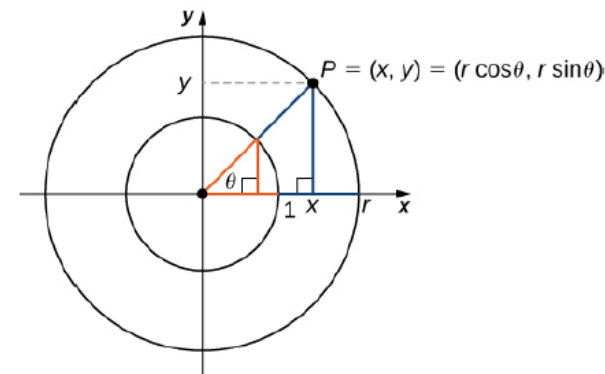
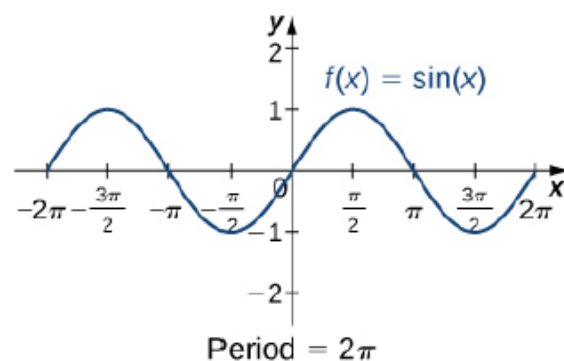
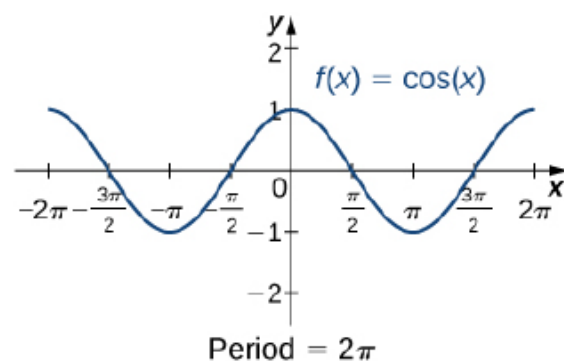


Figure 1.32

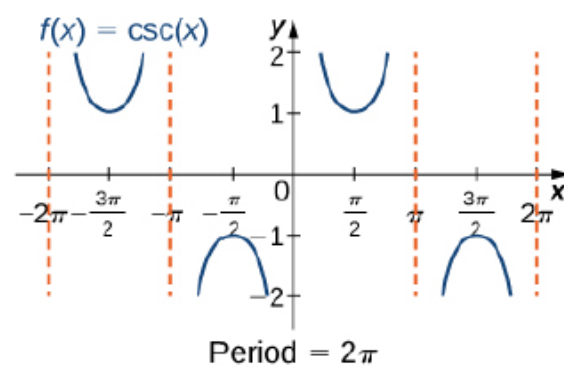
Figure 1.32 For a point $P = (x, y)$ on a circle of radius r , the coordinates x and y satisfy $x = r \cos \theta$ and $y = r \sin \theta$.



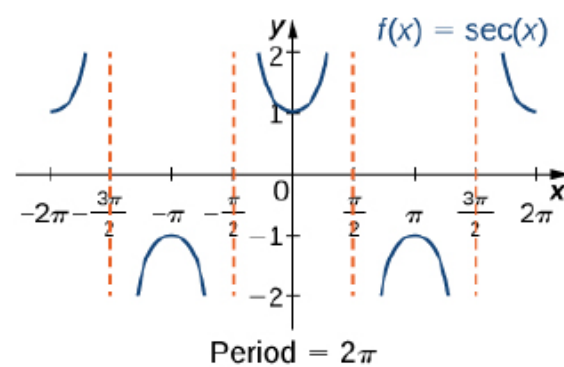
Period = 2π



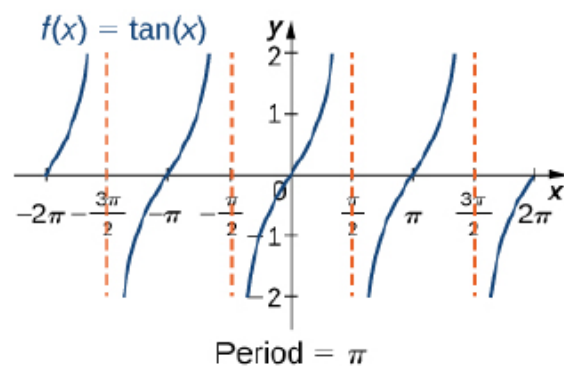
Period = 2π



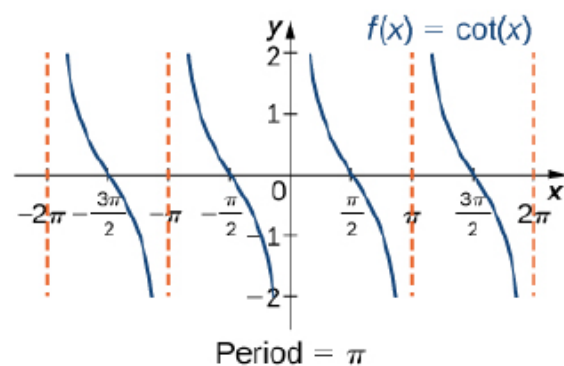
Period = 2π



Period = 2π



Period = π



Period = π

Figure 1.34 The six trigonometric functions are periodic.

The Inverse Function

Definition

Given a function f with domain D and range R , its **inverse function** (if it exists) is the function f^{-1} with domain R and range D such that $f^{-1}(y) = x$ if $f(x) = y$. In other words, for a function f and its inverse f^{-1} ,

$$f^{-1}(f(x)) = x \text{ for all } x \text{ in } D, \text{ and } f(f^{-1}(y)) = y \text{ for all } y \text{ in } R. \quad (1.11)$$

Note that f^{-1} is read as “ f inverse.” Here, the -1 is not used as an exponent and $f^{-1}(x) \neq 1/f(x)$. [Figure 1.37](#) shows the relationship between the domain and range of f and the domain and range of f^{-1} .

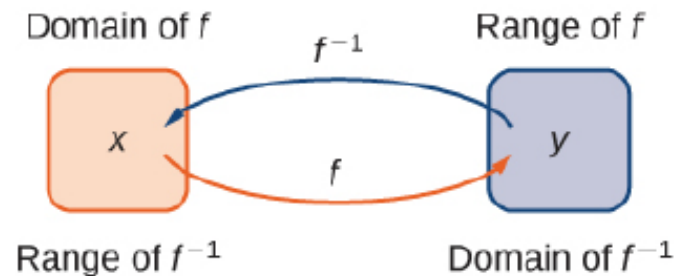


Figure 1.37 Given a function f and its inverse f^{-1} , $f^{-1}(y) = x$ if and only if $f(x) = y$. The range of f becomes the domain of f^{-1} and the domain of f becomes the range of f^{-1} .

One-to-One function

Recall that a function has exactly one output for each input. Therefore, to define an inverse function, we need to map each input to exactly one output. For example, let's try to find the inverse function for $f(x) = x^2$. Solving the equation $y = x^2$ for x , we arrive at the equation $x = \pm\sqrt{y}$. This equation does not describe x as a function of y because there are two solutions to this equation for every $y > 0$. The problem with trying to find an inverse function for $f(x) = x^2$ is that two inputs are sent to the same output for each output $y > 0$. The function $f(x) = x^3 + 4$ discussed earlier did not have this problem. For that function, each input was sent to a different output. A function that sends each input to a *different* output is called a one-to-one function.

Definition

We say a f is a **one-to-one function** if $f(x_1) \neq f(x_2)$ when $x_1 \neq x_2$.

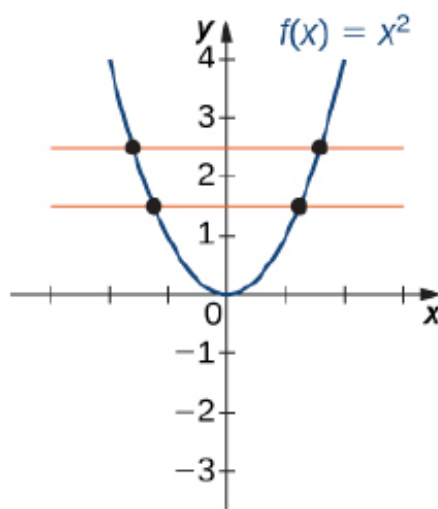
One way to determine whether a function is one-to-one is by looking at its graph. If a function is one-to-one, then no two inputs can be sent to the same output. Therefore, if we draw a horizontal line anywhere in the xy -plane, according to the **horizontal line test**, it cannot intersect the graph more than once. We note that the horizontal line test is different from the vertical line test. The vertical line test determines whether a graph is the graph of a function. The horizontal line test determines whether a function is one-to-one ([Figure 1.38](#)).

Rule: Horizontal Line Test

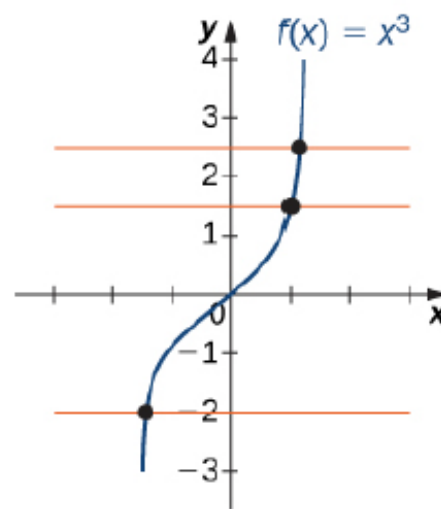
A function f is one-to-one if and only if every horizontal line intersects the graph of f no more than once.

In-Class activity

Are these functions one-to-one?



(a)



(b)

In-Class activity
Are these functions one-to-one?
solution

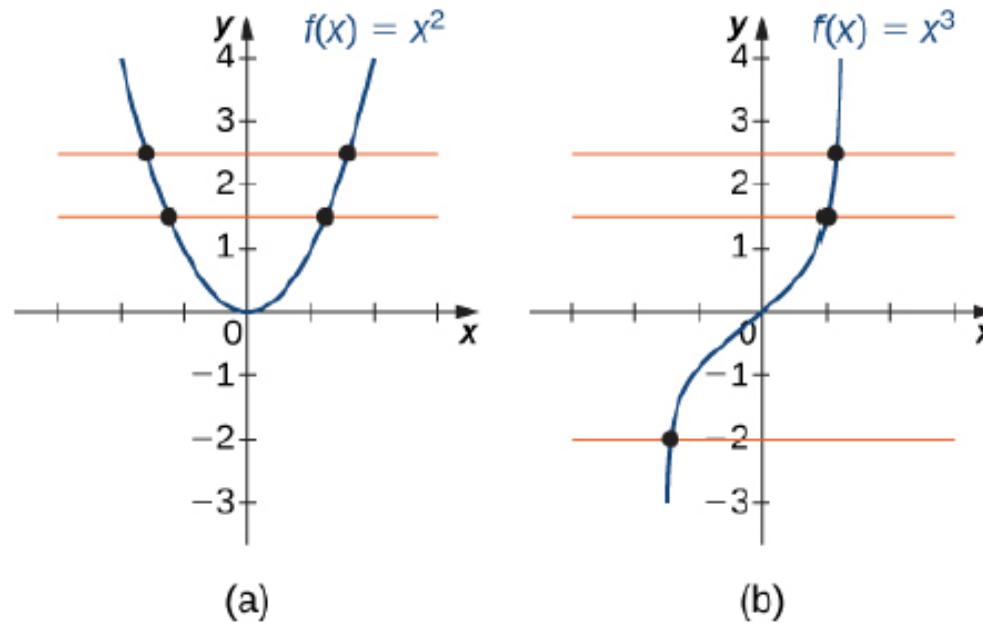
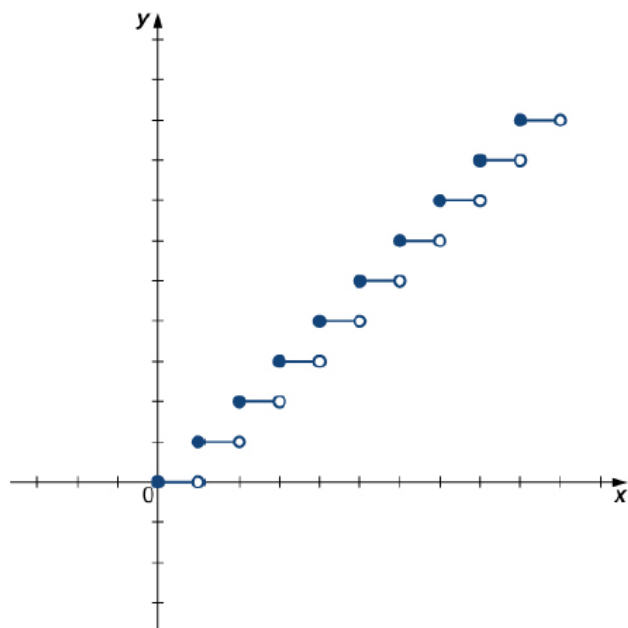


Figure 1.38 (a) The function $f(x) = x^2$ is not one-to-one because it fails the horizontal line test. (b) The function $f(x) = x^3$ is one-to-one because it passes the horizontal line test.

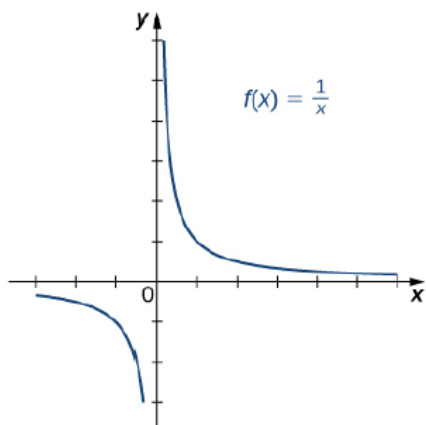
Determining Whether a Function Is One-to-One

For each of the following functions, use the horizontal line test to determine whether it is one-to-one.

a.



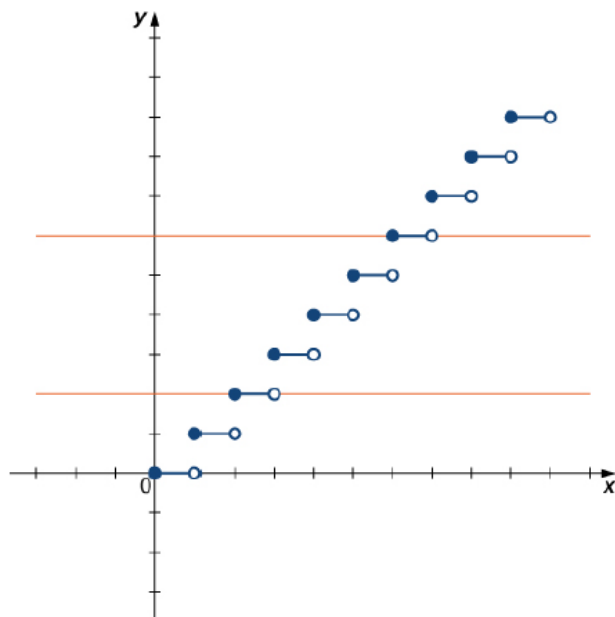
b.



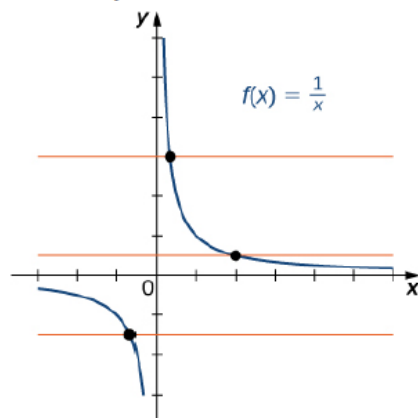
In-Class activity

✓ **Solution**

- a. Since the horizontal line $y = n$ for any integer $n \geq 0$ intersects the graph more than once, this function is not one-to-one.



- b. Since every horizontal line intersects the graph once (at most), this function is one-to-one.



In-Class activity
Solution (example 1.28)
study also example 1.29

Restricting domains

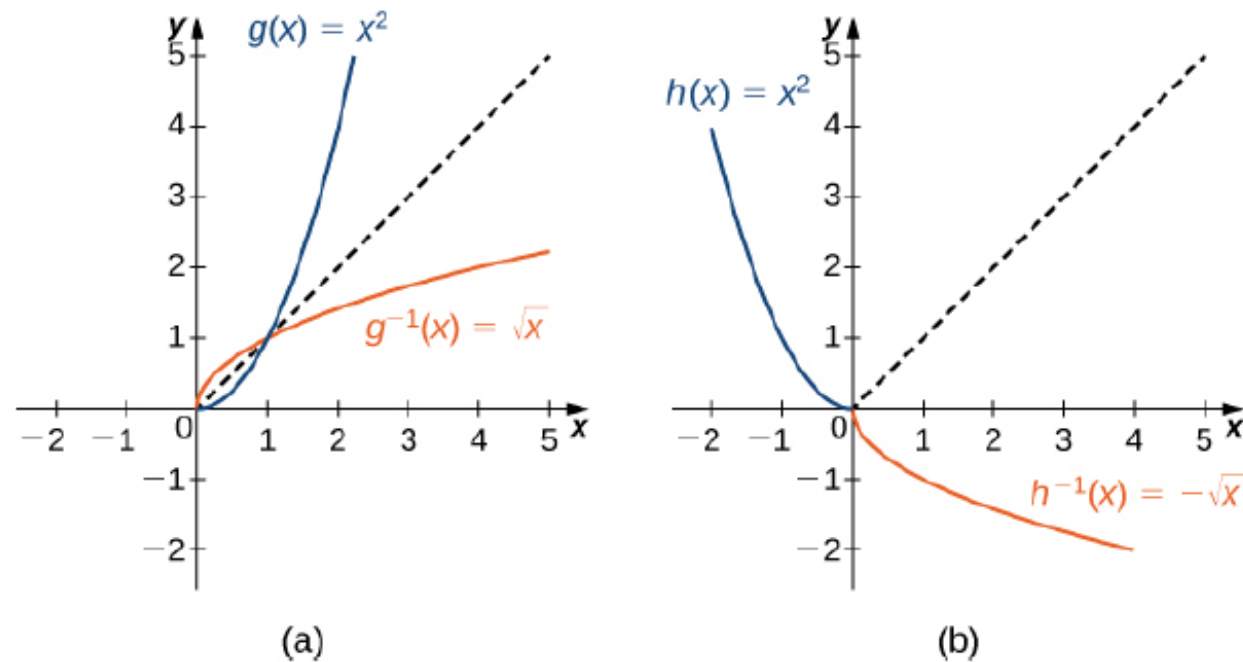


Figure 1.40 (a) For $g(x) = x^2$ restricted to $[0, \infty)$, $g^{-1}(x) = \sqrt{x}$. (b) For $h(x) = x^2$ restricted to $(-\infty, 0]$, $h^{-1}(x) = -\sqrt{x}$.

Exponential vs Power functions

Example

any function of the form $f(x) = b^x$, where $b > 0$, $b \neq 1$, is an

exponential function with **base** b and **exponent** x . Exponential functions have constant bases and variable exponents.

Note that a function of the form $f(x) = x^b$ for some constant b is not an exponential function but a power function.

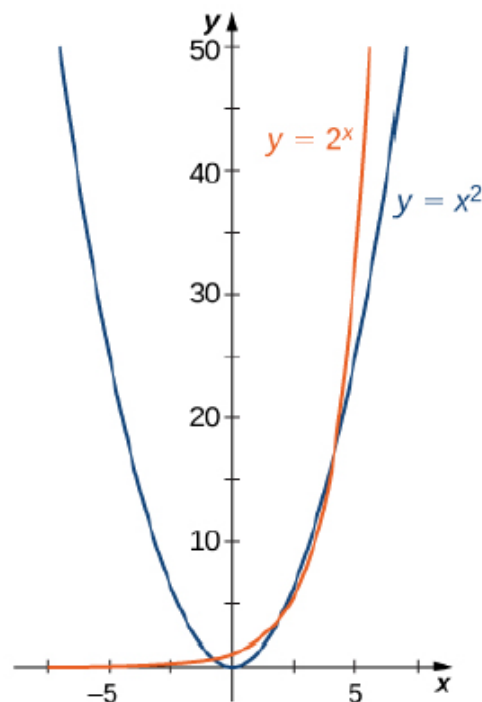


Figure 1.43 Both 2^x and x^2 approach infinity as $x \rightarrow \infty$, but 2^x grows more rapidly than x^2 . As $x \rightarrow -\infty$, $x^2 \rightarrow \infty$, whereas $2^x \rightarrow 0$.

In-Class activity

Rule: Laws of Exponents

For any constants $a > 0$, $b > 0$, and for all x and y ,

1. $b^x \cdot b^y = b^{x+y}$
2. $\frac{b^x}{b^y} = b^{x-y}$
3. $(b^x)^y = b^{xy}$
4. $(ab)^x = a^x b^x$
5. $\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$

Using the Laws of Exponents

Use the laws of exponents to simplify each of the following expressions.

- a. $\frac{(2x^{2/3})^3}{(4x^{-1/3})^2}$
- b. $\frac{(x^3 y^{-1})^2}{(xy^2)^{-2}}$

Solution

$$\frac{(2x^{2/3})^3}{(4x^{-1/3})^2} = \frac{2^3 (x^{2/3})^3}{4^2 (x^{-1/3})^2} = \frac{8x^2}{16x^{-2/3}} = \frac{x^2 x^{2/3}}{2} = \frac{x^{8/3}}{2}.$$

We can simplify as follows:

$$\frac{(x^3 y^{-1})^2}{(xy^2)^{-2}} = \frac{(x^3)^2 (y^{-1})^2}{x^{-2} (y^2)^{-2}} = \frac{x^6 y^{-2}}{x^{-2} y^{-4}} = x^6 x^2 y^{-2} y^4 = x^8 y^2.$$

Evaluating exponential functions

example

EXAMPLE 1.33

Bacterial Growth

Suppose a particular population of bacteria is known to double in size every 4 hours. If a culture starts with 1000 bacteria, the number of bacteria after 4 hours is $n(4) = 1000 \cdot 2$. The number of bacteria after 8 hours is $n(8) = n(4) \cdot 2 = 1000 \cdot 2^2$. In general, the number of bacteria after $4m$ hours is $n(4m) = 1000 \cdot 2^m$. Letting $t = 4m$, we see that the number of bacteria after t hours is $n(t) = 1000 \cdot 2^{t/4}$. Find the number of bacteria after 6 hours, 10 hours, and 24 hours.

✓ Solution

The number of bacteria after 6 hours is given by $n(6) = 1000 \cdot 2^{6/4} \approx 2828$ bacteria. The number of bacteria after 10 hours is given by $n(10) = 1000 \cdot 2^{10/4} \approx 5657$ bacteria. The number of bacteria after 24 hours is given by $n(24) = 1000 \cdot 2^6 = 64,000$ bacteria.

Logarithmic functions

The exponential function $f(x) = b^x$ is one-to-one, with domain $(-\infty, \infty)$ and range $(0, \infty)$. Therefore, it has an inverse function, called the *logarithmic function with base b* . For any $b > 0, b \neq 1$, the logarithmic function with base b , denoted $\log_b$, has domain $(0, \infty)$ and range $(-\infty, \infty)$, and satisfies

$$\log_b(x) = y \text{ if and only if } b^y = x.$$

For example,

$$\begin{array}{lll} \log_2(8) = 3 & \text{since} & 2^3 = 8, \\ \log_{10}\left(\frac{1}{100}\right) = -2 & \text{since} & 10^{-2} = \frac{1}{10^2} = \frac{1}{100}, \\ \log_b(1) = 0 & \text{since} & b^0 = 1 \text{ for any base } b > 0. \end{array}$$

Furthermore, since $y = \log_b(x)$ and $y = b^x$ are inverse functions,

$$\log_b(b^x) = x \text{ and } b^{\log_b(x)} = x.$$

The most commonly used logarithmic function is the function $\log_e$. Since this function uses natural e as its base, it is called the **natural logarithm**. Here we use the notation $\ln(x)$ or $\ln x$ to mean $\log_e(x)$. For example,

$$\ln(e) = \log_e(e) = 1, \ln(e^3) = \log_e(e^3) = 3, \ln(1) = \log_e(1) = 0.$$

Since the functions $f(x) = e^x$ and $g(x) = \ln(x)$ are inverses of each other,

$$\ln(e^x) = x \text{ and } e^{\ln x} = x,$$

Solving Equations Involving Logarithmic Functions

Solve each of the following equations for x .

- a. $\ln\left(\frac{1}{x}\right) = 4$
- b. $\log_{10}\sqrt{x} + \log_{10}x = 2$
- c. $\ln(2x) - 3\ln(x^2) = 0$

Solution

- a. By the definition of the natural logarithm function,

$$\ln\left(\frac{1}{x}\right) = 4 \text{ if and only if } e^4 = \frac{1}{x}.$$

Therefore, the solution is $x = 1/e^4$.

- b. Using the product and power properties of logarithmic functions, rewrite the left-hand side of the equation as

$$\log_{10}\sqrt{x} + \log_{10}x = \log_{10}x\sqrt{x} = \log_{10}x^{3/2} = \frac{3}{2}\log_{10}x.$$

Therefore, the equation can be rewritten as

$$\frac{3}{2}\log_{10}x = 2 \text{ or } \log_{10}x = \frac{4}{3}.$$

The solution is $x = 10^{4/3} = 10\sqrt[3]{10}$.

- c. Using the power property of logarithmic functions, we can rewrite the equation as $\ln(2x) - \ln(x^6) = 0$.
Using the quotient property, this becomes

$$\ln\left(\frac{2}{x^5}\right) = 0.$$

Therefore, $2/x^5 = 1$, which implies $x = \sqrt[5]{2}$. We should then check for any extraneous solutions.

Chapter Opener: The Richter Scale for Earthquakes



Figure 1.48 (credit: modification of work by Robb Hannawacker, NPS)

In 1935, Charles Richter developed a scale (now known as the *Richter scale*) to measure the magnitude of an earthquake. The scale is a base-10 logarithmic scale, and it can be described as follows: Consider one earthquake with magnitude R_1 on the Richter scale and a second earthquake with magnitude R_2 on the Richter scale. Suppose $R_1 > R_2$, which means the earthquake of magnitude R_1 is stronger, but how much stronger is it than the other earthquake? A way of measuring the intensity of an earthquake is by using a seismograph to measure the amplitude of the earthquake waves. If A_1 is the amplitude measured for the first earthquake and A_2 is the amplitude measured for the second earthquake, then the amplitudes and magnitudes of the two earthquakes satisfy the following equation:

$$R_1 - R_2 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Consider an earthquake that measures 8 on the Richter scale and an earthquake that measures 7 on the Richter scale. Then,

$$8 - 7 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Therefore,

$$\log_{10} \left(\frac{A_1}{A_2} \right) = 1,$$

which implies $A_1/A_2 = 10$ or $A_1 = 10A_2$. Since A_1 is 10 times the size of A_2 , we say that the first earthquake is 10 times as intense as the second earthquake. On the other hand, if one earthquake measures 8 on the Richter scale and another measures 6, then the relative intensity of the two earthquakes satisfies the equation

$$\log_{10} \left(\frac{A_1}{A_2} \right) = 8 - 6 = 2.$$

Therefore, $A_1 = 100A_2$. That is, the first earthquake is 100 times more intense than the second earthquake.

How can we use logarithmic functions to compare the relative severity of the magnitude 9 earthquake in Japan in 2011 with the magnitude 7.3 earthquake in Haiti in 2010?

Example 1.39 (page 88)

Solution

To compare the Japan and Haiti earthquakes, we can use an equation presented earlier:

$$9 - 7.3 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Therefore, $A_1/A_2 = 10^{1.7}$, and we conclude that the earthquake in Japan was approximately 50 times more intense than the earthquake in Haiti.

✓ **Solution**

To compare the Japan and Haiti earthquakes, we can use an equation presented earlier:

$$9 - 7.3 = \log_{10} \left(\frac{A_1}{A_2} \right).$$

Therefore, $A_1/A_2 = 10^{1.7}$, and we conclude that the earthquake in Japan was approximately 50 times more intense than the earthquake in Haiti.

Reminder!

At the end of each Chapter, the textbook (*OpenStax Calculus I*) offers:

Chapter Review

- Key terms
- Key Equations
- Key Concepts
- Review Exercises

Chapter Review

Key Terms

absolute value function $f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

algebraic function a function involving any combination of only the basic operations of addition, subtraction, multiplication, division, powers, and roots applied to an input variable x

base the number b in the exponential function $f(x) = b^x$ and the logarithmic function $f(x) = \log_b x$

composite function given two functions f and g , a new function, denoted $g \circ f$, such that $(g \circ f)(x) = g(f(x))$

cubic function a polynomial of degree 3; that is, a function of the form $f(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$

decreasing on the interval I a function decreasing on the interval I if, for all $x_1, x_2 \in I$, $f(x_1) \geq f(x_2)$ if $x_1 < x_2$

degree for a polynomial function, the value of the largest exponent of any term

dependent variable the output variable for a function

domain the set of inputs for a function

even function a function is even if $f(-x) = f(x)$ for all x in the domain of f

exponent the value x in the expression b^x

function a set of inputs, a set of outputs, and a rule for mapping each input to exactly one output

graph of a function the set of points (x, y) such that x is in the domain of f and $y = f(x)$

horizontal line test a function f is one-to-one if and only if every horizontal line intersects the graph of f , at most, once

hyperbolic functions the functions denoted $\sinh$, $\cosh$, $\tanh$, csch , sech , and coth , which involve certain combinations of e^x and e^{-x}

increasing on the interval I a function increasing on the interval I if for all $x_1, x_2 \in I$, $f(x_1) \leq f(x_2)$ if $x_1 < x_2$

independent variable the input variable for a function

inverse function for a function f , the inverse function f^{-1} satisfies $f^{-1}(y) = x$ if $f(x) = y$

inverse hyperbolic functions the inverses of the hyperbolic functions where $\cosh$ and sech are restricted to the domain $[0, \infty)$; each of these functions can be expressed in terms of a composition of the natural logarithm function and an algebraic function

inverse trigonometric functions the inverses of the trigonometric functions are defined on restricted domains where they are one-to-one functions

linear function a function that can be written in the form $f(x) = mx + b$

logarithmic function a function of the form $f(x) = \log_b(x)$ for some base $b > 0$, $b \neq 1$ such that $y = \log_b(x)$ if and only if $b^y = x$

mathematical model A method of simulating real-life situations with mathematical equations

natural exponential function the function $f(x) = e^x$

natural logarithm the function $\ln x = \log_e x$

number e as m gets larger, the quantity $(1 + (1/m))^m$ gets closer to some real number; we define that real number to be e ; the value of e is approximately 2.718282

odd function a function is odd if $f(-x) = -f(x)$ for all x in the domain of f

one-to-one function a function f is one-to-one if $f(x_1) \neq f(x_2)$ if $x_1 \neq x_2$

Key Terms

Key Equations

Key Concepts

Key Equations

Composition of two functions	$(g \circ f)(x) = g(f(x))$
Absolute value function	$f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$
Point-slope equation of a line	$y - y_1 = m(x - x_1)$
Slope-intercept form of a line	$y = mx + b$
Standard form of a line	$ax + by = c$
Polynomial function	$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$
Generalized sine function	$f(x) = A \sin(B(x - \alpha)) + C$
Inverse functions	$f^{-1}(f(x)) = x$ for all x in D , and $f(f^{-1}(y)) = y$ for all y in R .

Key Concepts

1.1 Review of Functions

- A function is a mapping from a set of inputs to a set of outputs with exactly one output for each input.
- If no domain is stated for a function $y = f(x)$, the domain is considered to be the set of all real numbers x for which the function is defined.
- When sketching the graph of a function f , each vertical line may intersect the graph, at most, once.
- A function may have any number of zeros, but it has, at most, one y -intercept.
- To define the composition $g \circ f$, the range of f must be contained in the domain of g .
- Even functions are symmetric about the y -axis whereas odd functions are symmetric about the origin.

1.2 Basic Classes of Functions

Review Exercises

to better prepare for Tests/Exams focus on the:
Lecture examples,
Practical session exercises,
Independent Study Assignments

Review Exercises

True or False? Justify your answer with a proof or a counterexample.

310. A function is always one-to-one.
311. $f \circ g = g \circ f$, assuming f and g are functions.
312. A relation that passes the horizontal and vertical line tests is a one-to-one function.
313. A relation passing the horizontal line test is a function.

For the following problems, state the domain and range of the given functions:

$$f = x^2 + 2x - 3, \quad g = \ln(x - 5), \quad h = \frac{1}{x+4}$$

314. h
315. g
316. $h \circ f$
317. $g \circ f$

Find the degree, y-intercept, and zeros for the following polynomial functions.

318. $f(x) = 2x^2 + 9x - 5$
319. $f(x) = x^3 + 2x^2 - 2x$

Simplify the following trigonometric expressions.

320. $\frac{\tan^2 x}{\sec^2 x} + \cos^2 x$
321. $\cos^2 x - \sin^2 x$